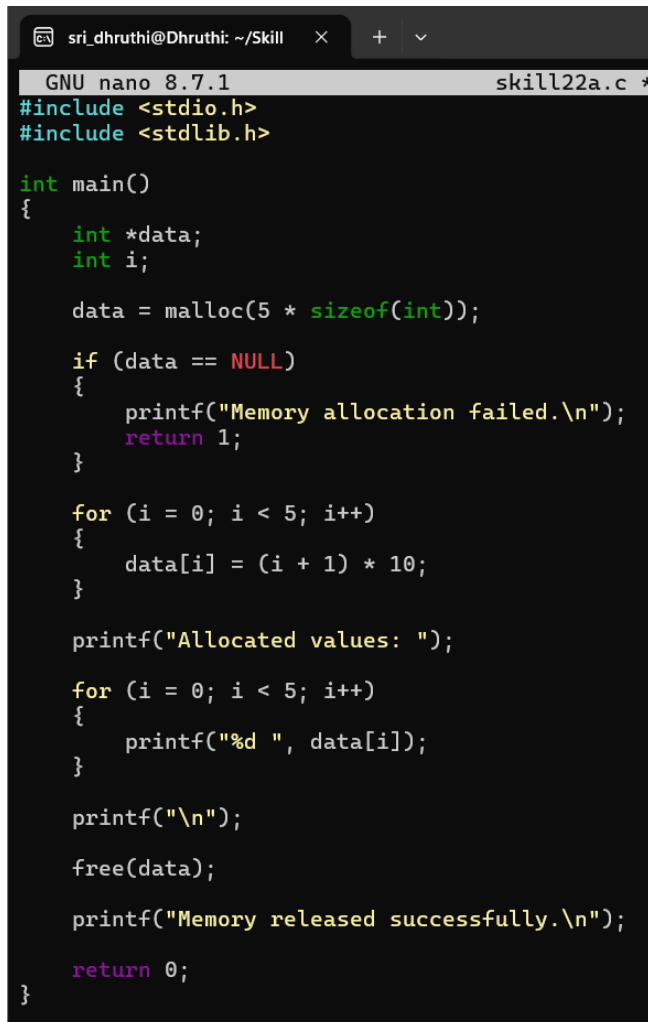


Os skill 22

22(a) Valgrind – Detect and Fix Memory Leaks

What it does: Allocates memory, uses it, releases it with free(), and allows Valgrind to verify that there are no memory leaks.



```
GNU nano 8.7.1 skill22a.c *
#include <stdio.h>
#include <stdlib.h>

int main()
{
    int *data;
    int i;

    data = malloc(5 * sizeof(int));

    if (data == NULL)
    {
        printf("Memory allocation failed.\n");
        return 1;
    }

    for (i = 0; i < 5; i++)
    {
        data[i] = (i + 1) * 10;
    }

    printf("Allocated values: ");

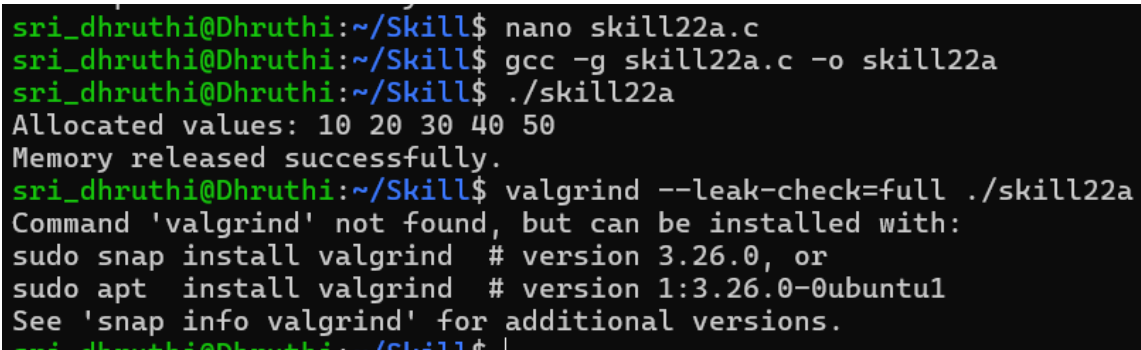
    for (i = 0; i < 5; i++)
    {
        printf("%d ", data[i]);
    }

    printf("\n");

    free(data);

    printf("Memory released successfully.\n");

    return 0;
}
```



```
sri_dhruthi@Dhruthi:~/Skill$ nano skill22a.c
sri_dhruthi@Dhruthi:~/Skill$ gcc -g skill22a.c -o skill22a
sri_dhruthi@Dhruthi:~/Skill$ ./skill22a
Allocated values: 10 20 30 40 50
Memory released successfully.
sri_dhruthi@Dhruthi:~/Skill$ valgrind --leak-check=full ./skill22a
Command 'valgrind' not found, but can be installed with:
sudo snap install valgrind # version 3.26.0, or
sudo apt install valgrind # version 1:3.26.0-0ubuntu1
See 'snap info valgrind' for additional versions.
```

22(b) Test Cases and Automated Testing

What it does: Creates simple test cases, compares expected results, and generates a test report.

```
sri_dhruthi@Dhruthi: ~/Skill × + v
GNU nano 8.7.1 skill22b.c
#include <stdio.h>

int main()
{
    int passed = 0;
    int total = 3;

    /* Test Case 1 */
    if (2 + 2 == 4)
    {
        printf("Test 1: PASS\n");
        passed++;
    }
    else
    {
        printf("Test 1: FAIL\n");
    }

    /* Test Case 2 */
    if (5 * 3 == 15)
    {
        printf("Test 2: PASS\n");
        passed++;
    }
    else
    {
        printf("Test 2: FAIL\n");
    }

    /* Test Case 3 */
    if (10 > 3)
    {
        printf("Test 3: PASS\n");
        passed++;
    }
    else
    {

```

```
sri_dhruthi@Dhruthi:~/Skill$ nano skill22b.c
sri_dhruthi@Dhruthi:~/Skill$ gcc skill22b.c -o skill22b
sri_dhruthi@Dhruthi:~/Skill$ ./skill22b
Test 1: PASS
Test 2: PASS
Test 3: PASS

Test Report
-----
Passed: 3
Failed: 0
Total : 3
Overall Result: PASS
```